Mohammad Shamsi

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Address:
Sharif University of Technology, Tehran, Iran

Education

M.Sc. — Structural Engineering, Sharif University of Technology, Tehran, Iran 2024-present
Current GPA: 17.16 out of 20
B.Sc. — Civil Engineering, Shahid Bahonar University of Kerman, Kerman, Iran 2020-2024
GPA: 17.36 out of 20

Research Interests

- Structural Health Monitoring
- Random Vibration
- Machine Learning & Deep Learning
- Earthquake Engineering & RC Wall Behavior
- Finite Element Method
- Vulnerability & Seismic Retrofitting of Structures

Academic Awards

- Bronze Medal (3rd Place) in Civil Engineering Olympiad, Iran National Olympiad, 2024**.**
- Member of Iran's National Foundation.
- Ranked 2nd among 66 Peers in B.Sc. Civil Engineering, 2024.
- Member of the Civil Engineering Scientific Association at Shahid Bahonar University of Kerman, 2022.

Teaching Experience

- Teaching Assistant, *Dynamics*, Shahid Bahonar University — Fall 2023 & Spring 2024
- Teaching Assistant, *Concrete Structures I*, Sharif University of Technology — Spring 2025
- Teaching Assistant, *Physics I*, Sharif University of Technology—Fall 2025

Selected Projects

- Fall 2025 – Random Vibration of Structures (Graduate Course Project, Grade: 18.1/20):
  - Experimental modal identification of multi-degree-of-freedom structural systems under sine-sweep and random excitation.
  - Estimation of natural frequencies, damping ratios, mode shapes, stationarity, and ergodicity of stochastic excitation records.
  - Frequency-domain and time-domain analysis of structural response under impact and random loading using PSD and FRF methods.
  - Stochastic response analysis of fixed-base and base-isolated (HDRB/LRB) RC frame structures subjected to white-noise and spectrum-compatible earthquake excitations.
  - Computation of response statistics, spectral density functions, impulse response functions, and soil-filtered seismic inputs using Python.
- Spring 2025 – Micro–Nano Mechanics: Q-Factor Prediction of Comb-Drive MEMS Actuators using analytical and numerical approaches.
- Fall 2025 – Review Paper: “Comprehensive Review of Experimental Studies on U-Shaped and Rectangular RC Walls under Quasi-Static Cyclic Loading.”
- Spring 2024 – Structural Modeling and Design: Analysis and design of reinforced concrete and steel structures using ETABS, SAP2000, and SAFE.

Technical Skills

Programming Language: PYTHON – MATLAB

Civil software: SAP2000 – ETABS – SAFE – Opensees– SesmoStruct

Engineering Software: Abaqus – Comsol Multiphysics

Working Experience

Structural Engineer (Internship): Tahkim Sazeh Khour, Jun - Sep 2024

Structural Engineer : Abar Fanavari Energy

Language Skills

- **Persian:** National Language Proficiency
- **English:** Upper Intermediate (CEFR: B2)

Conference Papers & Presentations

• Shamsi, M., and Bakhshi, A. (2026).
"Hybrid Physics-Informed Neural Networks Integrating Finite Element and Sensor Data for Structural Health Monitoring and Digital Twin Development."
Accepted for Oral Presentation at the International Conference on Noise and Vibration Engineering (ISMA 2026) and the International Conference on Uncertainty in Structural Dynamics (USD 2026), KU Leuven, Leuven, Belgium, September 7–9, 2026.

• Full conference paper submitted to the ISMA 2026–USD 2026 Conference Proceedings (publication expected in 2026).

Research Experience

M.Sc. Thesis, Sharif University of Technology (Ongoing)

Hybrid Physics-Informed Neural Networks Integrating Finite Element and Sensor Data for Structural Health Monitoring and Digital Twin Development.

Supervisor: Dr. Bakhshi

• Development of Physics-Informed Neural Networks (PINNs) for structural response prediction and damage detection.
• Integration of finite element simulations and sensor measurements for digital twin development.
• Application of machine learning and structural dynamics methods for Structural Health Monitoring.